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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
IV Semester B. E. Examinations October-2023
Industrial Engineering and Management
CAD/CAM AND ROBOTICS

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

1	1.1	Define Robot Programming.	02
	1.2	List any two advantages of suction grippers.	02
	1.3	Identify any two objectives of an Industrial Robot.	02
	1.4	If given an opportunity to build a robot to be used in the department of IEM at RVCE, what problem would the robot solve and what would be its functionality?	02
	1.5	_____ and _____ are special purpose tool which are used to facilitate production (machining, assembling and inspection operations) when work piece is based on the concept of interchangeability according to which every part will be produced within an established tolerance.	02
	1.6	Students who carry out productivity studies at industries as part of major project sometimes tend to conclude that one or set of robots can be used to automate the entire process and enhance productivity by multifold and cost of commissioning it can be recovered in a very short period by compensating the salaries of the employee who can be removed from the process. Review such inference from the standpoint of India as a country.	02
	1.7	Identify two disadvantages of automation.	02
	1.8	Identify two applications of robots in industrial sector and service sector.	02
	1.9	Tactile sensors are also called as _____ and are sensitive to _____.	02
	1.10	State the Zeroth Law of robotics.	02

PART-B

2	a	Explain the Shigley model with a block diagram.	10
	b	Enumerate any six benefits of CAD in automation industry.	06
3	a	Write a CNC peck drilling program for the profile given in Fig 3a. The cutting tool used is slot drill cutter and the depth of cut is 5mm.	

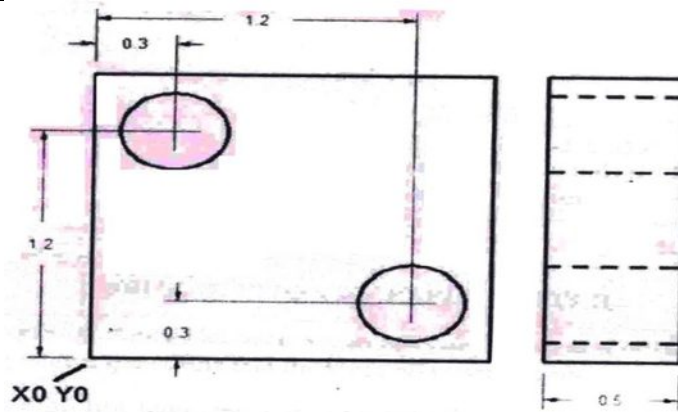


Fig 3a

	b	Discuss the various elements of a typical Numerical Control System. OR	10 06	
4	a	Explain with the help of an example i) Linear interpolation ii) Circular interpolation.	06	
	b	Explain any five types of Distributed Numerical Control Systems.	10	
5	a	Identify major socio-economic considerations favoring automation.	08	
	b	With example, state the difference between mechanization and automation. OR	08	
6	a	Explain how low-cost automation is achieved in manufacturing industry.	10	
	b	Describe six major issues for Automation in Factory Operations.	06	
7	a	The robot has to do the following functions. Recommend the types of sensors it should be fitted with to carry out the operations successfully. Also mention one specification of the recommended sensor. Functionality 1: A binary output signal when it comes in contact with an object. Functionality 2: To sense and indicate presence of any object without physical contact. Functionality 3: To identify the distance of sensor from the object. Functionality 4: To estimate the speed with which a manipulator is moved.	08	
	b	Explain the advantages and disadvantages of vacuum/suction grippers. OR	08	
8	a	Explain the characteristics of an industrial robot.	10	
	b	How are the robots classified? Explain.	06	
9	a	Assume that the robot must pick up objects from a chute and place them in successive boxes. Choose the sequence of robot activity for which robot programming will have to be written at a later stage.	08	
	b	Explain any two robot programming techniques. OR	08	
10	a	Explain the factors considered for selection of gripper.	10	
	b	Write short notes on types of sensors in robots.	06	